STAT528 Competition1

Wang, Yeting Wu, [illegible] Li

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1 Introduction

Sentiment classification of movie reviews is a widely used benchmark task in natural language processing and has practical value for understanding user opinions at scale. In this project, we study binary sentiment classification on the IMDB movie review dataset, which contains balanced positive and negative labels. Although the dataset is clean and well-structured, movie reviews exhibit variable lengths and context-dependent language, making sentiment modeling non-trivial. We conduct exploratory data analysis, apply a unified preprocessing pipeline, and fine-tune two pre-trained transformer models, DistilBERT and RoBERTa, under the same training and evaluation settings. By comparing their performance using standard classification metrics, this study evaluates the practical effectiveness and trade-offs of transformer-based models for sentiment classification.

2 Exploratory Data Analysis

2.1 Data Overview

The dataset comprises 25,000 samples with 3 features. The data structure is clean with no missing values and no duplicate entries. Each sample contains a unique identifier, the review text, and a binary sentiment label.

2.2 Class Distribution Visualization

The dataset contains 25,000 reviews with an equal number of positive (label=1) and negative (label=0) samples. As shown in Figure 1, the class distribution is well balanced, which allows reliable evaluation using accuracy and F1 score.

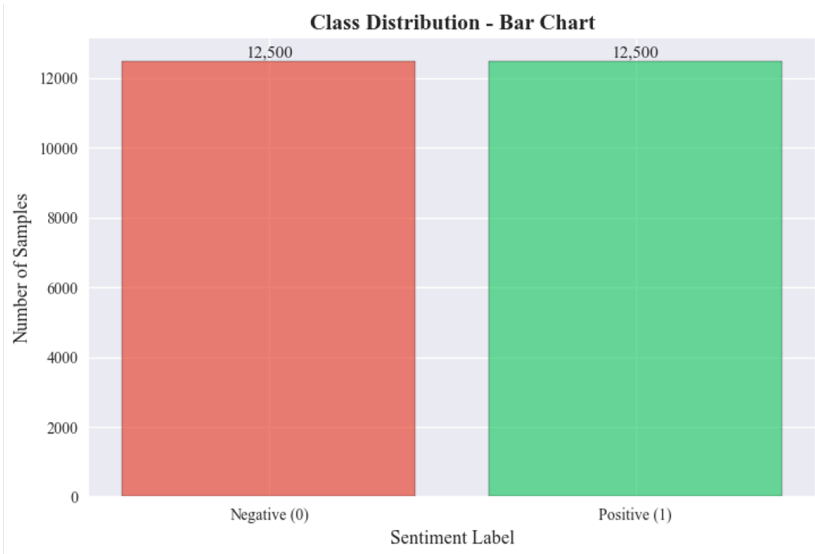


Figure 1: Class distribution of IMDB review sentiment labels

2.3 Review Length Analysis

Review length is an important feature that may correlate with sentiment expression. This analysis examines both character count and word count distributions across different sentiment classes.

2.3.1 Length Statistics

We compute descriptive statistics for review lengths

Table 1: Review length statistics by sentiment label.

Label	Character Length					Word Count				
	Mean	Median	Std	Min	Max	Mean	Median	Std	Min	Max
0	1302.98	976.50	957.07	52	8969	230.87	174.00	166.66	10	1522
1	1347.16	982.00	1046.75	70	13704	236.71	174.00	180.49	12	2470

The average review contains 1325 characters, with a median length of 979 characters. In terms of word counts, reviews have an average of 234 words and a median of 174 words. The two sentiment classes exhibit similar length distributions, suggesting that review length alone is unlikely to be a strong discriminator between labels.

2.3.2 Length Distribution Visualization

These visualizations help identify whether there are systematic differences in review length between sentiment classes, which could be informative features for the classification model.

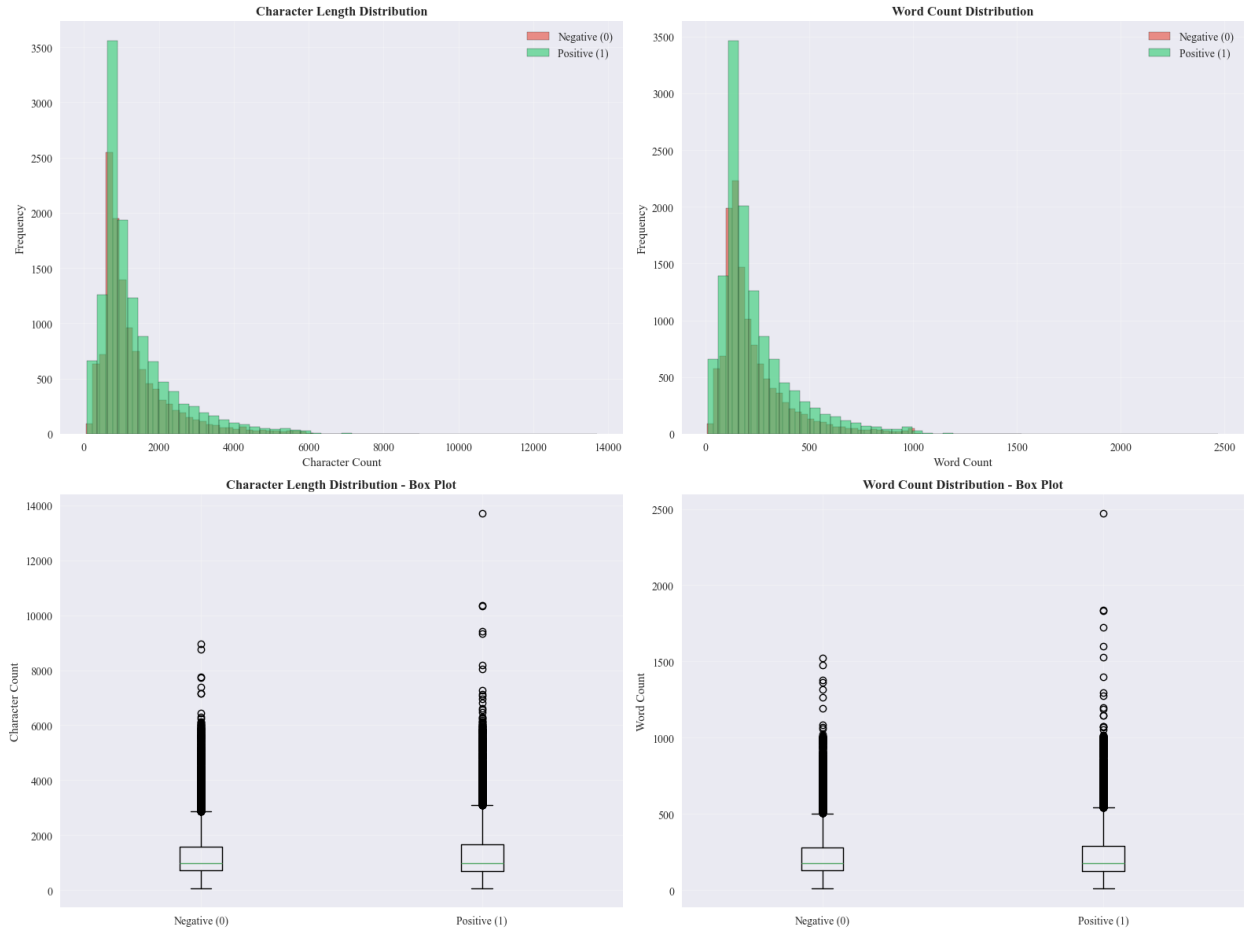


Figure 2: Character and word length distributions by sentiment class

2.4 Vocabulary and Text Content

Understanding the vocabulary and word usage patterns is essential for feature engineering and model development. This analysis examines frequent words and their associations with different sentiment classes.

2.4.1 Frequent Words Analysis

We identify the most frequent words in both positive and negative reviews.

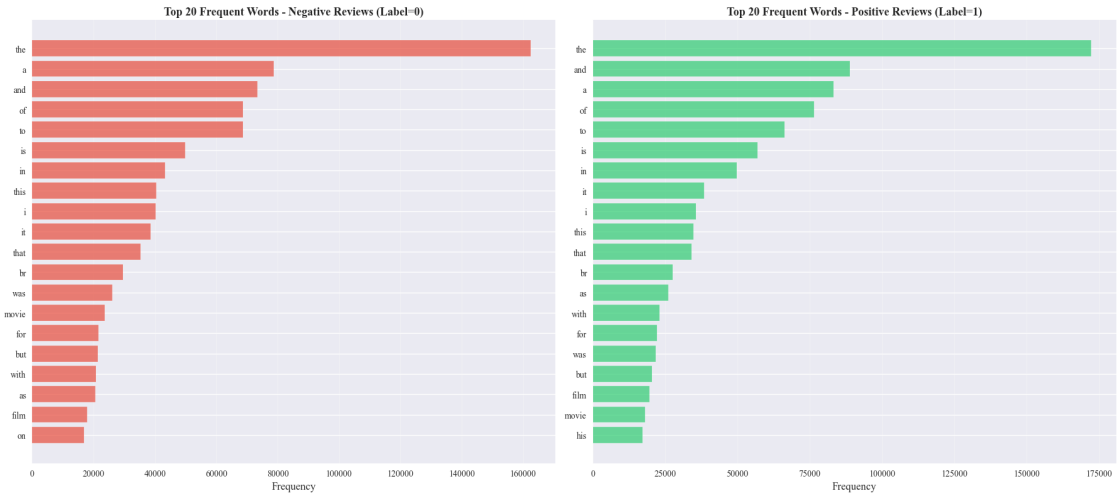


Figure 3: Top frequent words in negative and positive reviews

2.4.2 Word-Sentiment Associations

To identify words that are strongly associated with specific sentiments, we computed normalized word frequencies for each class and calculated difference scores indicating words that appear more frequently in one class than the other. The analysis identified a ranked list of words with high sentiment discriminative power.

The analysis reveals that words with high discriminative power (difference score > 0.9) include terms that appear significantly more often in one sentiment class than the other. Table 2 presents the top 20 words with the highest discriminative power, ranked by their difference scores.

It is worth noting that many of the highly discriminative words identified in Table 2 are movie-specific entities, such as character names, titles, or numeric references. These terms reflect dataset-specific sentiment cues rather than general sentiment vocabulary, highlighting the importance of contextual understanding in sentiment classification. Words such as "410", "boll", "210", and "uwe" appear almost exclusively in negative reviews (with negative class proportions above 99%), while words like "paulie", "iturbi", "gundam", and "philo" are strongly associated with positive reviews (with positive class proportions above 97%). These highly discriminative words can serve as important features for sentiment classification, as they provide strong signals for distinguishing between positive and negative sentiment classes.

2.5 Sample Reviews and Qualitative Inspection

Qualitative inspection of sample reviews provides insight into the characteristics of the dataset and helps validate labeling quality. By examining randomly selected reviews from each sentiment class, we assess writing style, content diversity, and the consistency between textual content and assigned sentiment labels.

Reviews vary substantially in both length and style. Negative reviews often contain detailed critiques of plot structure, character development, and overall film quality. For example, sample ID 3450 (2,008 characters, 356 words) presents an extensive negative evaluation of multiple aspects of the film. In contrast, positive reviews tend to emphasize specific elements that resonated with the viewer, such as character motivations or narrative themes, as illustrated by sample ID 3605 (613 characters, 119 words).

Table 2: Top 20 Words with High Sentiment Discriminative Power

Word	Neg. Freq.	Pos. Freq.	Neg. Prop.	Pos. Prop.	Diff.
410	0.000056	0.000000	99.40%	0.60%	0.9880
boll	0.000043	0.000000	99.23%	0.77%	0.9845
210	0.000038	0.000000	99.11%	0.89%	0.9821
uwe	0.000035	0.000000	99.03%	0.97%	0.9807
paulie	0.000000	0.000034	1.02%	98.98%	0.9797
thunderbirds	0.000021	0.000000	98.40%	1.60%	0.9680
beowulf	0.000019	0.000000	98.29%	1.71%	0.9658
awfulbr	0.000016	0.000000	97.92%	2.08%	0.9585
wayans	0.000016	0.000000	97.92%	2.08%	0.9585
ajay	0.000015	0.000000	97.83%	2.17%	0.9566
seagal	0.000046	0.000001	97.82%	2.18%	0.9563
iturbi	0.000000	0.000015	2.23%	97.77%	0.9554
dahmer	0.000015	0.000000	97.73%	2.27%	0.9546
010	0.000015	0.000000	97.73%	2.27%	0.9546
gundam	0.000001	0.000029	2.33%	97.67%	0.9534
philo	0.000000	0.000014	2.38%	97.62%	0.9523
giovanna	0.000000	0.000014	2.50%	97.50%	0.9500
lindy	0.000000	0.000014	2.50%	97.50%	0.9500
feinstone	0.000000	0.000014	2.50%	97.50%	0.9500
steaming	0.000013	0.000000	97.43%	2.57%	0.9486

Extremely short reviews (character count < 100) convey sentiment concisely, such as the statement "A rating of '1' does not begin to express how dull, depressing and relentlessly bad this movie is" (98 characters, label 0). Very long reviews (character count > 2000) typically include detailed thematic discussions and extended analysis. Overall, this qualitative inspection reveals no systematic labeling issues and highlights patterns that may influence model behavior during training.

2.6 Data Artifacts and Cleaning Needs

Raw text data contains several artifacts and formatting issues that may affect model performance. The most prevalent artifact is HTML tags and markup, which appear in 14,667 reviews (58.67% of the dataset), most commonly in the form of `<br />` line breaks. These tags do not convey sentiment information and can introduce noise if left unprocessed.

URLs and email addresses occur infrequently, with 61 reviews (0.24%) containing web links and 23 reviews (0.09%) containing email addresses. While rare, these elements are generally irrelevant to sentiment classification and should be removed or replaced with placeholders during preprocessing. In addition, 60 reviews (0.24%) contain formatting irregularities such as multiple consecutive spaces or tabs, which motivates whitespace normalization.

Data quality checks indicate no empty or extremely short reviews (character count ≤ 5), suggesting good overall data completeness. However, 96 reviews (0.38%) contain duplicated text, which may lead to data leakage if not addressed. Based on these findings, preprocessing steps including HTML tag removal, normalization of whitespace, handling of URLs and email addresses, and deduplication are applied to improve data quality and ensure reliable model training.

3 Methodology

This section describes the methodology employed for sentiment classification, including data preprocessing and feature engineering, model implementations, and training, validation, and hyperparameter tuning procedures. Two distinct pre-trained language models are implemented and compared: DistilBERT (Model 1) and RoBERTa (Model 2).

3.1 Data Preprocessing & Feature Engineering

Both models employ an identical preprocessing pipeline to ensure a fair comparison. Raw review text is cleaned by removing HTML tags using regular expressions (e.g., `re.sub(r'<[^>]+>', ' ', text)`), motivated by the high prevalence of markup artifacts observed during exploratory analysis. The text is then lowercased, non-alphanumeric characters are removed, and consecutive whitespace is normalized to produce a standardized textual representation.

The cleaned text is tokenized using the corresponding pretrained tokenizer (DistilBERT tokenizer for Model 1 and RoBERTa tokenizer for Model 2). Tokenization produces subword representations with truncation and padding applied to a maximum sequence length of 256 tokens. The resulting `input_ids` and `attention_mask` are used as model inputs.

The dataset is split into training and validation sets using stratified random sampling with a 90/10 ratio to preserve class balance, and a fixed random seed of 42 is used to ensure reproducibility.

3.2 Models to Implement

3.2.1 Model 1: DistilBERT

Model 1 implements DistilBERT (Distilled Bidirectional Encoder Representations from Transformers), a lighter and faster variant of BERT that retains 97% of BERT's performance while being 60% faster. The model uses the `distilbert-base-uncased` pre-trained checkpoint from Hugging Face Transformers.

DistilBERT is a transformer-based architecture that employs a distilled version of BERT's architecture, using 6 transformer layers instead of BERT's 12 layers. The model is fine-tuned for binary sentiment classification by adding a classification head on top of the pre-trained encoder. The classification head consists of a linear layer that maps the [CLS] token representation to two output logits corresponding to negative (0) and positive (1) sentiment classes.

The model architecture leverages the pre-trained knowledge from large-scale language modeling tasks, allowing it to understand contextual relationships in text. During fine-tuning, all model parameters are updated, enabling the model to adapt its representations specifically for the sentiment classification task.

3.2.2 Model 2: RoBERTa

Model 2 implements RoBERTa (Robustly Optimized BERT Pretraining Approach), an improved variant of BERT that achieves better performance through optimized training procedures. The model uses the `roberta-base` pre-trained checkpoint from Hugging Face Transformers.

RoBERTa maintains BERT's architecture (12 transformer layers) but improves upon BERT through several key modifications: removing the next sentence prediction objective, training on longer sequences, using dynamic masking, and training with larger batches and learning rates. These optimizations result in superior performance on downstream tasks, particularly in sentiment analysis.

Similar to Model 1, RoBERTa is fine-tuned for binary sentiment classification by adding a classification head. The model processes input sequences and produces logits for the two sentiment classes. RoBERTa's enhanced pre-training makes it particularly effective at capturing nuanced sentiment expressions in movie reviews.

3.3 Training, Validation, and Tuning

3.3.1 Training Configuration

Both models are trained using similar optimization strategies with some key differences. Model 1 (DistilBERT) uses a learning rate of 2×10^{-5} with 10 training epochs, while Model 2 (RoBERTa) uses a learning rate of 1×10^{-5} with 20 training epochs. Both models employ the AdamW optimizer with a weight decay of 0.01 to prevent overfitting.

Both models use a linear learning rate schedule with warmup. Model 1 supports optional warmup steps (default 0, meaning no warmup), while Model 2 uses a warmup ratio of 0.1, meaning 10% of the total training steps are used for learning rate warmup. The warmup strategy gradually increases the learning rate from 0 to the target learning rate, which helps stabilize training in the early epochs.

Gradient clipping is applied to both models with a maximum gradient norm of 1.0, preventing exploding gradients during training. Model 1 additionally supports gradient accumulation, allowing effective larger batch sizes by accumulating gradients over multiple steps before updating parameters.

3.3.2 Validation and Model Selection

Both models use the same validation strategy. During training, each epoch evaluates the model on the validation set, computing accuracy, precision, recall, and F1 score. The best model is selected based on validation F1 score, as F1 score provides a balanced measure of precision and recall for binary classification tasks.

Model checkpoints are saved whenever the validation F1 score improves. For Model 1, checkpoints are saved after each epoch to enable training resumption, while the best model state is maintained separately. Model 2 saves only the best model based on F1 score. Both models load the best checkpoint for final evaluation and test set prediction.

3.3.3 Hyperparameter Settings

The key hyperparameters for both models are summarized as follows. Both models use a maximum sequence length of 256 tokens, batch size of 16, and weight decay of 0.01. Model 1 uses a learning rate of 2×10^{-5} for 10 epochs, while Model 2 uses a learning rate of 1×10^{-5} for 20 epochs. The validation split ratio is 0.1 (10%) for both models, and the random seed is set to 42 for reproducibility.

4 Results

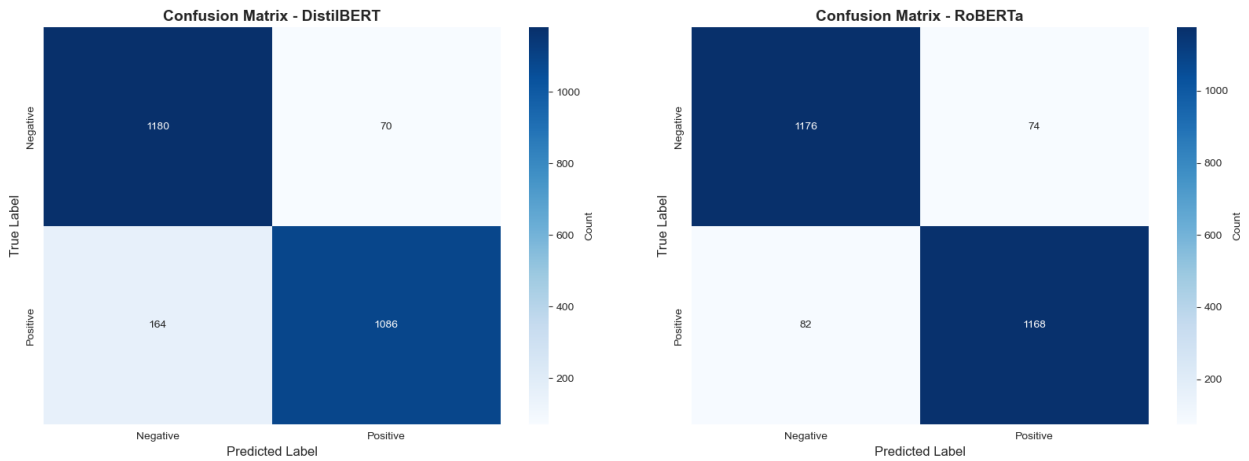
This section presents the experimental results for both models on the validation set. The models were evaluated using standard classification metrics: accuracy, precision, recall, and F1 score. The validation set consists of 2,500 samples (10% of the training data), with balanced class distribution maintained through stratified sampling.

4.1 Model 1: DistilBERT Results

Model 1 (DistilBERT) was trained for 10 epochs with a learning rate of 2×10^{-5}. The model achieved its best validation F1 score of 0.9122 at epoch 7. The final evaluation on the validation set yielded the following performance metrics:

The metrics indicate a trade-off between precision and recall. While the high precision reflects a conservative prediction strategy that minimizes false positives, the comparatively lower recall suggests the model tends to miss a portion of positive reviews. The confusion matrix in Figure 4A further illustrates this distribution. Validation performance peaked at epoch 7 (F1: 0.9122), with subsequent epochs showing signs of minor overfitting.

Figure 4A shows the confusion matrix for Model 1, providing detailed insight into the classification performance across both sentiment classes.



(A) DistilBERT **(B)** RoBERTa

Figure 4: Confusion matrices for DistilBERT (A) and RoBERTa (B) on the validation set.

4.2 Model 2: RoBERTa Results

Model 2 (RoBERTa) was trained for 20 epochs with a learning rate of 1×10^{-5} and a warmup ratio of 0.1. The model achieved its best validation F1 score of 0.9370 at epoch 13. The final evaluation on the validation set yielded the following performance metrics:

RoBERTa demonstrates superior performance across all metrics, achieving an excellent balance between precision and recall. Notably, the model improves recall by 6.56% over DistilBERT without compromising precision. This robustness results in a 3.43% increase in the F1 score and a total accuracy of 93.72%, attributing to the model's enhanced capacity to capture nuanced sentiment expressions.

Figure 4B presents the confusion matrix for Model 2, allowing for detailed analysis of classification errors.

4.3 Performance Summary

Table 3 provides a comprehensive comparison of both models' performance metrics on the validation set.

Table 3: Model Performance Comparison on Validation Set

Model	Accuracy	Precision	Recall	F1 Score
DistilBERT (Model 1)	0.9064	0.9394	0.8688	0.9027
RoBERTa (Model 2)	0.9372	0.9397	0.9344	0.9370

Both models were evaluated on the same stratified validation set of 2,500 samples (10% of the data) to ensure a fair comparison.

5 Discussion

5.1 Model Comparison

The comparison between DistilBERT and RoBERTa reveals important insights about the trade-offs between model efficiency and performance in sentiment classification tasks.

5.1.1 Architectural Differences

DistilBERT employs a distilled architecture with 6 transformer layers, making it approximately 60% faster and 40% smaller than BERT while retaining 97% of BERT's performance. This design makes DistilBERT particularly suitable for scenarios requiring faster inference or limited computational resources. In contrast, RoBERTa maintains the full 12-layer architecture of BERT but benefits from improved pre-training procedures, including dynamic masking, longer sequences, and removal of the next sentence prediction objective.

5.1.2 Performance Characteristics

Based on the experimental results, DistilBERT achieved a validation F1 score of 0.9027 with high precision (0.9394) but moderate recall (0.8688). This precision-recall trade-off suggests that DistilBERT is conservative in predicting positive sentiment, prioritizing correctness over coverage. The model's high precision makes it particularly valuable in applications where false positives are costly.

RoBERTa demonstrates superior performance with a validation F1 score of 0.9370, achieving excellent balance between precision (0.9397) and recall (0.9344). Unlike DistilBERT, RoBERTa maintains high precision while significantly improving recall, resulting in a more balanced and robust classifier. The 3.43 percentage point improvement in F1 score (from 0.9027 to 0.9370) demonstrates the benefits of RoBERTa's enhanced pre-training procedures and larger model capacity. RoBERTa's improved recall (0.9344 vs. 0.8688) indicates better coverage of positive sentiment cases, making it more suitable for applications requiring comprehensive sentiment detection.

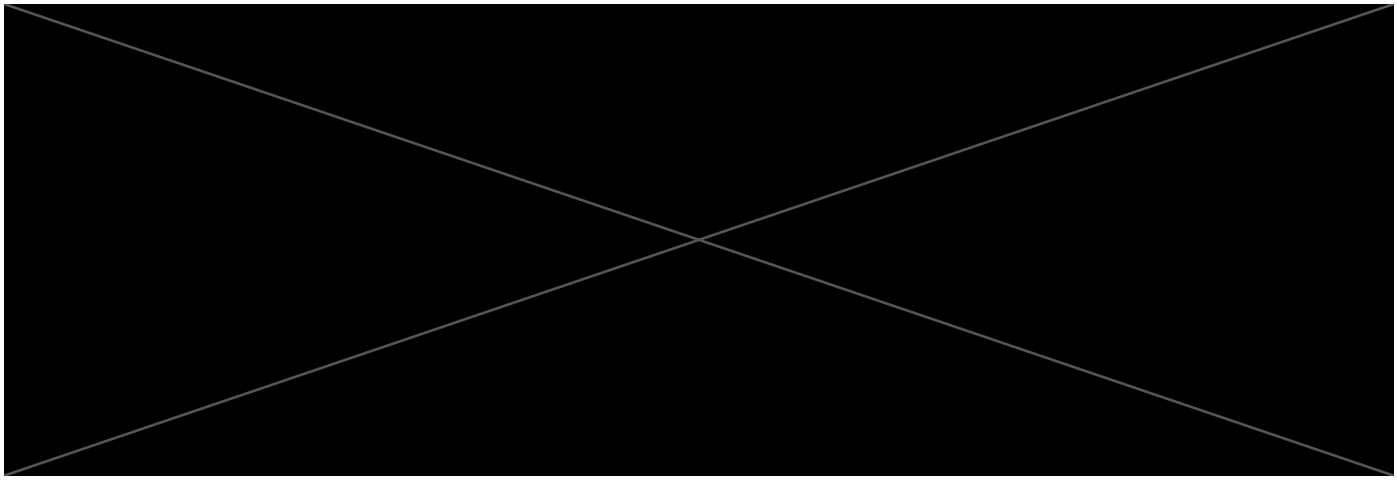
5.1.3 Training Efficiency

Model 1 (DistilBERT) required 10 epochs to reach optimal performance, with the best model found at epoch 7. The training process showed signs of overfitting after epoch 7, as indicated by decreasing validation F1 scores despite continued training loss reduction. Model 2 (RoBERTa) was trained for 20 epochs with a lower learning rate (1×10^{-5} vs. 2×10^{-5}), reflecting the need for more careful fine-tuning of the larger model. RoBERTa achieved its best performance at epoch 13 (F1: 0.9370), demonstrating more stable training with consistent improvements over a longer training period. The model maintained strong performance through epoch 20, indicating good generalization without severe overfitting.

5.1.4 Computational Considerations

DistilBERT's smaller architecture (66.9 million parameters) offers advantages in terms of memory usage and inference speed, making it more suitable for deployment in resource-constrained environments. RoBERTa's larger model size provides greater representational capacity but requires more computational resources for both training and inference.

5.2 Limitations and Future Work

This study has several limitations that should be acknowledged. First, the evaluation is limited to two transformer-based models (DistilBERT and RoBERTa), and a more comprehensive comparison could include other architectures such as BERT, ALBERT, or ELECTRA to provide broader insights into model selection. Second, the models are evaluated on a single movie review dataset, and their generalization to other domains (e.g., product reviews, social media posts) or languages remains unverified. Third, hyperparameters were selected based on common practices rather than systematic optimization through grid search or Bayesian optimization, which could potentially improve performance. Additionally, both models truncate sequences to 256 tokens, potentially losing information from longer reviews, and the black-box nature of transformer models limits interpretability of their decision-making process.

Future research directions include several promising avenues for improvement. Ensemble methods combining predictions from multiple models could potentially improve overall performance through model diversity. More sophisticated preprocessing techniques such as spell correction, handling of slang and abbreviations, or domain-specific preprocessing could enhance model performance. Data augmentation techniques including back-translation, synonym replacement, or paraphrasing could increase training data diversity and improve model robustness. Additionally, exploring domain-specific transfer learning, multi-task learning approaches, active learning strategies, model optimization techniques for deployment, and cross-lingual capabilities would contribute to both improving model performance and expanding the applicability of sentiment classification systems to broader domains and use cases.